

# The Christoffel–Darboux Kernel as a Pandrosion Cross-Quotient

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## Abstract

The Christoffel–Darboux kernel of an orthogonal polynomial family  $\{P_n\}$  can be expressed as a *Pandrosion cross-quotient*:  $K_n(x, y) = \gamma_n (P_n(y) Q_{P_{n+1}}(y, x) - P_{n+1}(y) Q_{P_n}(y, x))$ , where  $Q_f(a, z) = (f(z) - f(a))/(z - a)$ . The confluent form gives the Wronskian  $W(P_n, P_{n+1})$  via Pandrosion self-values  $Q(x, x) = f'(x)$ . Gauss quadrature weights become  $w_k = 2/((1 - \xi_k^2) Q_{P_n}(\xi_k, \xi_k)^2)$ , and the zero interlacing property is a sign alternation condition on the Pandrosion quotients. All results are verified for Legendre and Hermite polynomials.

## 1 The Pandrosion cross-quotient

**Definition 1.1** (Cross-quotient). Given two polynomials  $f, g$  and points  $x, y$  with  $x \neq y$ :

$$\mathcal{C}(f, g; x, y) := \frac{f(x)g(y) - g(x)f(y)}{x - y}. \quad (1)$$

**Lemma 1.2.** *The cross-quotient decomposes into Pandrosion quotients:*

$$\boxed{\mathcal{C}(f, g; x, y) = g(y) Q_f(y, x) - f(y) Q_g(y, x),} \quad (2)$$

where  $Q_f(a, z) = (f(z) - f(a))/(z - a)$ .

*Proof.*  $\mathcal{C} = (f(x)g(y) - f(y)g(x) + f(y)g(y) - g(x)f(y))/(x - y) = g(y)(f(x) - f(y))/(x - y) - f(y)(g(x) - g(y))/(x - y)$ .  $\square$

*Remark 1.3.* When  $f = g$ ,  $\mathcal{C}(f, f; x, y) = 0$  (antisymmetry). When  $g = 1$ ,  $\mathcal{C}(f, 1; x, y) = Q_f(y, x)$ : the cross-quotient reduces to the standard Pandrosion quotient.

## 2 The Christoffel–Darboux kernel

Let  $\{P_n\}_{n \geq 0}$  be an orthogonal polynomial family with respect to a weight  $w$  on  $(a, b)$ , satisfying the three-term recurrence  $P_{n+1}(x) = (A_n x + B_n)P_n(x) - C_n P_{n-1}(x)$ .

**Theorem 2.1** (Christoffel–Darboux = Pandrosion cross-quotient). *The reproducing kernel satisfies:*

$$\boxed{K_n(x, y) = \gamma_n \mathcal{C}(P_{n+1}, P_n; x, y) = \gamma_n (P_n(y) Q_{P_{n+1}}(y, x) - P_{n+1}(y) Q_{P_n}(y, x))} \quad (3)$$

for an explicit constant  $\gamma_n = A_n / \|P_n\|^2$ .

*Proof.* This is the classical Christoffel–Darboux formula rewritten via Lemma 1.2. The factor  $(P_{n+1}(x)P_n(y) - P_n(x)P_{n+1}(y))/(x - y)$  is  $\mathcal{C}(P_{n+1}, P_n; x, y)$  by definition.  $\square$

### 3 Confluent form and the Wronskian

**Corollary 3.1** (Confluent kernel). *As  $y \rightarrow x$ :*

$$K_n(x, x) = \gamma_n (P'_{n+1}(x) P_n(x) - P'_n(x) P_{n+1}(x)) = \gamma_n W(P_n, P_{n+1})(x), \quad (4)$$

where the derivatives are Pandrosion self-values:  $P'_n(x) = Q_{P_n}(x, x)$ .

The confluent kernel is the Wronskian of consecutive orthogonal polynomials, computed via Pandrosion self-evaluation.

### 4 Gauss quadrature via Pandrosion

**Theorem 4.1** (Pandrosion quadrature weights). *The Gauss quadrature weights at the zeros  $\xi_1, \dots, \xi_n$  of  $P_n$  are:*

$$w_k = \frac{2}{(1 - \xi_k^2) Q_{P_n}(\xi_k, \xi_k)^2}. \quad (5)$$

*Proof.* The classical formula uses  $P'_n(\xi_k)$ , which equals  $Q_{P_n}(\xi_k, \xi_k)$  by definition of the Pandrosion self-value.  $\square$

**Numerical verification (Gauss–Legendre,  $n = 5$ ):**

| $k$ | $\xi_k$   | $Q_{P_5}(\xi_k, \xi_k)$ | $w_k$ (Pandrosion) | $w_k$ (exact) |
|-----|-----------|-------------------------|--------------------|---------------|
| 1   | −0.906180 | 6.8703                  | 0.23692689         | 0.23692689    |
| 2   | −0.538469 | −2.4259                 | 0.47862867         | 0.47862867    |
| 3   | 0.000000  | 1.8750                  | 0.56888889         | 0.56888889    |
| 4   | 0.538469  | −2.4259                 | 0.47862867         | 0.47862867    |
| 5   | 0.906180  | 6.8703                  | 0.23692689         | 0.23692689    |

### 5 Interlacing via Pandrosion signs

**Proposition 5.1** (Interlacing = sign alternation). *The zeros of  $P_n$  and  $P_{n+1}$  interlace if and only if  $P_{n+1}(\xi_k)$  alternates sign at the zeros  $\xi_k$  of  $P_n$ . Equivalently:*

$$\text{sgn } P_{n+1}(\xi_k) = (-1)^{n-k} \quad (k = 1, \dots, n). \quad (6)$$

This is the Pandrosion sign condition: the evaluation of ‘the next polynomial’ at ‘the current zeros’ must alternate.

### 6 Summary: complete dictionary

| Orthogonal polynomials                                                      | Pandrosion form                                                          |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------------|
| CD kernel $K_n(x, y)$                                                       | $\gamma_n \mathcal{C}(P_{n+1}, P_n; x, y)$                               |
| Confluent $K_n(x, x)$                                                       | $\gamma_n W(P_n, P_{n+1})$ via $Q(x, x)$                                 |
| Gauss weights $w_k$                                                         | $2/(1 - \xi_k^2) \cdot Q_{P_n}(\xi_k, \xi_k)^{-2}$                       |
| Zero interlacing                                                            | $\text{sgn } P_{n+1}(\xi_k) = (-1)^{n-k}$                                |
| Three-term recurrence                                                       | $Q_{P_{n+1}}(\alpha_n, x) = P_n(x) - \beta_n^2 Q_{P_{n-1}}(\alpha_n, x)$ |
| Verified: Legendre ( $n = 2, \dots, 7$ ) and Hermite ( $n = 2, \dots, 7$ ). |                                                                          |

The Christoffel–Darboux kernel—the central object of approximation theory, spectral theory, and random matrix universality—is a Pandrosion cross-quotient of consecutive orthogonal polynomials.

## References

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